Modeling in Quantitative Finance



Matt Merino
Perry Fan
MSBA 502 Analytics



Setting The Stage

What is quant trading?
Using mathematical models to find market anomalies and build predictive trading signals

What do quants actually do?
Analyzes vast alternative datasets (sentiment, volume, macro stats, tick data) using complex models in Python

Why we are researching this
Machine learning is “more robust than its predecessors but hardly revolutionary”
– Chief Scientist at Man AHL, Anthony Ledford

Choosing A Target

Returns

-Returns are mostly noise
(Gu, Kelly and Xiu, 2020)

-Our initial test agreed:
Every model scored a negative R2.

-Overfitting risk:
RF Train R2 (.85), Test (-.17)

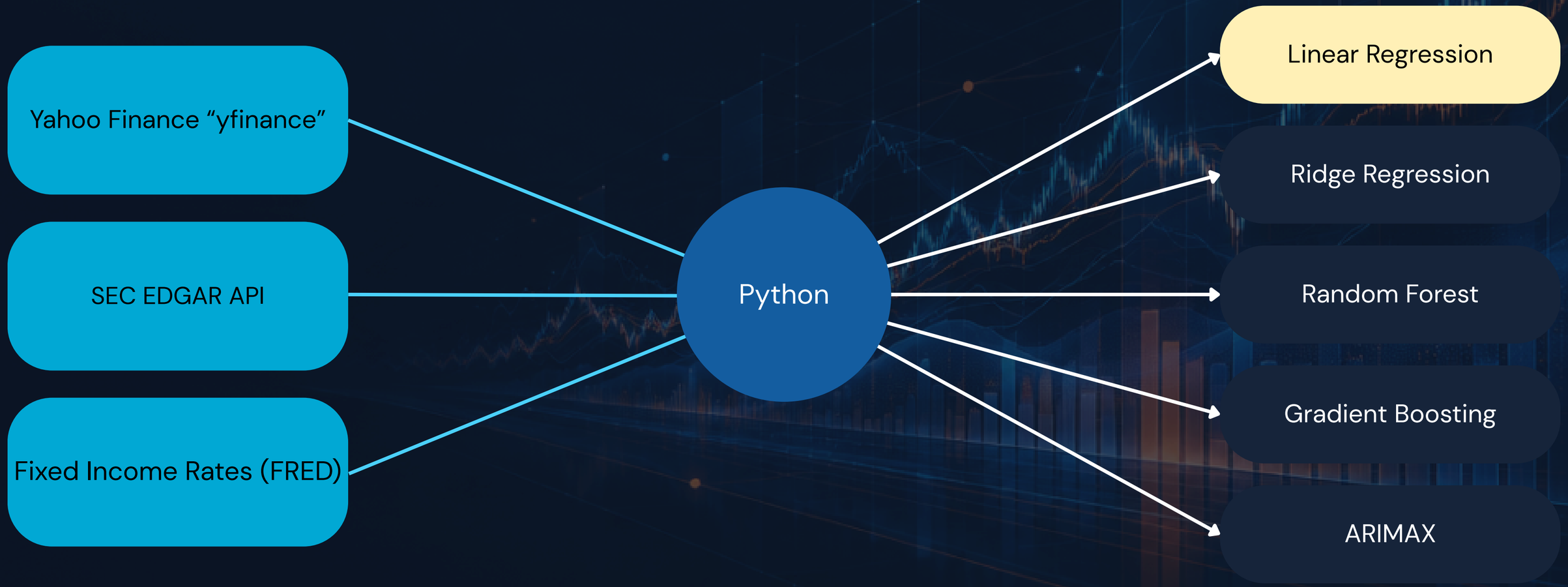
Volatility

-Volatility is predictable

-Easily explainable to a client

-Key part of portfolio performance metrics: Sharpe

The Data & Models



The Variables

log_vol_1w	log_vol_1m	log_vol_3m	log_idio_vol_1m
log_vol_12m	beta_12m	ret_1m	mom_12_1
ma_gap_200	dd_52w	max_ret_1m	log_dollar_vol
log_vix	tbill_chg_12m	curve_chg_12m	credit
log_spy_vol_1m	log_mcap	ep	bm
leverage	roa	cash_assets	ocf_assets
sales_growth	loss	fund_missing	earn_next
sec_Consumer_Discretionary	sec_Consumer_Staples	sec_Energy	sec_Financials
sec_Health_Care	sec_Industrials	sec_Information_Technology	sec_Materials
sec_Real_Estate	sec_Utilities		

Set A

Set B

Analysis - Test Data

	Set A (12 Variables)	Set B (All 39 variables)
Linear Regression	0.52 | 8.4	0.27 | 10.9
Ridge Regression	0.52 | 8.4	0.46 | 9.0
Random Forest	0.522 | 8.3	0.528 | 8.5
Gradient Boosted	0.525 | 8.3	0.44 | 9.5
ARIMAX	0.43 | 9.1	0.45 | 9.0

The Winners vs Baseline

Naive Baseline

Last months volatility

R^2 = 0.29

Avg Error = 10.9

Complexity: None

Volatility Index (VIX) * Stock Riskiness

Market Estimate of next months S&P 500 volatility * Stocks individual Riskiness

R^2 = 0.39

Avg Error = 10.6

Complexity: Very Low

VIX × (stock's 12-month volatility ÷ S&P 500's 12-month volatility)

Random Forest (All Variables)

R^2 = 0.525

Avg Error = 8.3

Complexity: High

Linear Regression (12 variable)

R^2 = 0.52

Avg Error = 8.4

Complexity: Low

Takeaways & Conclusions

1. **Returns are unpredictable.** Every model scored at or below 0% R^2 on unseen data.
2. **Risk is about half predictable**. The best models explained about 52% of next month's volatility across 495 stocks.
3. **Simple matched complex**. OLS on 12 price variables (52.0%) came within 1 point of random forest (52.8%).
4. **More variables can hurt**. Adding 27 variables dropped plain OLS from 52% to 27% when interest rates jumped in 2022 and 2023.
5. **We beat the VIX.** By 13 points for individual stocks; for the S&P 500, a simple regression matched or beat the options market's forecast.

- **Survivorship bias**: only today's S&P 500 members, so companies that failed or left the index are excluded.
- **One regime dominates the test**: 2019 to 2026 was mostly a tech-led bull market.
- **Monthly data**: professionals use intraday data to measure volatility more precisely.
- **Market-level VIX test:** only 93 months, so the edge over the VIX is small.
- **Earnings flag is approximate:** based on SEC filing dates, not exact announcement dates.

Conclusion

Do investors need complex quant models?

No. Quants often actually use linear regression, and our data shows why: it was as accurate as machine learning, easier to explain, and harder to break when markets changed.

What this means

- *Retail investors*: Be skeptical of products that claim to predict returns. No model we tested could.
- *Financial advisors*: Use volatility forecasts to set client expectations and size positions.
- *Risk managers & funds*: A simple, transparent model on price history forecasts risk as well as machine learning and holds up better when the economy shifts.

Next steps

- Intraday data for more precise volatility
- Stock-level options data (each company's own "VIX")
- Include delisted companies to remove survivorship bias Thank you! Questions?